DWSIM - Task 3 Screening Task - FOSSEE

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Surrogate Modeling of a Binary Distillation Column Using DWSIM and Machine Learning

System: Benzene–Toluene | **Thermodynamic Model:** Peng–Robinson EOS | **Simulator:** DWSIM (Synthetic Mode)

1. Problem Statement:

Binary distillation is one of the most energy-intensive and widely deployed separation processes in the chemical and petrochemical industries, typically accounting for 40–60% of total plant energy consumption. Process simulators such as DWSIM implement rigorous tray-by-tray solutions to the MESH equations — Material balance, Equilibrium, Summation, and Heat balance — at each theoretical stage. While these simulations are physically accurate, they are computationally expensive to execute repeatedly. Each individual call to the simulator involves iterative convergence over a system of nonlinear algebraic equations, making large-scale parametric studies, real-time optimisation, uncertainty quantification, and digital twin integration effectively infeasible through direct simulation.

A surrogate model, also referred to as a metamodel or

response surface, resolves this bottleneck. It is a data-driven statistical approximation trained on a library of simulation outputs that maps any valid set of operating conditions to the corresponding column performance indicators instantly, without invoking the underlying simulator. The surrogate trades a small, quantifiable loss in accuracy for a reduction in computation time of several orders of magnitude — from minutes per simulation call to microseconds per prediction.

This report documents the complete end-to-end development of a machine learning surrogate model for a Benzene–Toluene binary distillation column. The surrogate predicts four key performance indicators — distillate purity (xD), bottoms purity (xB), condenser duty (QC), and reboiler duty (QR) — from seven independently controllable operating conditions. Four machine learning architectures are trained, evaluated, and compared on a held-out test set, and a final model is selected with justification based on accuracy, generalisability, and physical consistency.

2. System Description and DWSIM Flowsheet:

2.1 Binary Mixture Selection

The Benzene–Toluene system was selected from the permitted mixture list. This is the canonical test case for binary distillation in chemical engineering education and research. Benzene (normal boiling point 80.1°C at 1 atm) is the more volatile component; Toluene (110.6°C) is the less volatile. The relative volatility α of benzene over toluene

varies from approximately 2.13 to 2.52 across the operating conditions studied, computed as a function of temperature and pressure using the Peng–Robinson equation of state. This moderate and well-behaved relative volatility produces a system that is neither trivially easy nor intractably difficult to separate, making it well-suited as a surrogate modelling benchmark.

2.2 Thermodynamic Model

The Peng–Robinson (PR) cubic equation of state was selected as the thermodynamic framework. PR EOS is the industry-standard model for vapour–liquid equilibrium in light hydrocarbon and aromatic systems, providing reliable predictions of fugacity coefficients, enthalpy departures, and vapour-phase compressibility without the need for liquid-phase activity coefficient correlations. For the Benzene–Toluene system at the pressures and temperatures studied (1.0–3.0 atm, 60–120°C), PR EOS reproduces experimentally measured VLE data to within standard engineering tolerances and is the recommended model in DWSIM for this class of system.

2.3 DWSIM Column Configuration

The distillation column is configured in DWSIM as a rigorous equilibrium-stage distillation block. The key configuration assumptions are as follows. A total condenser is assumed, producing a saturated liquid distillate. A partial reboiler is used and is counted as one theoretical equilibrium stage. The column pressure is assumed equal to the feed pressure throughout (no tray pressure drop).

The inside-out algorithm is used for tray-by-tray convergence, which is the recommended solver for binary and multicomponent distillation in DWSIM due to its robust handling of composition-dependent K-values.

The base case operating configuration is: total stages N = 20, feed stage NF = 10 (counting from the top), reflux ratio RR = 2.5, feed mole fraction of benzene zF = 0.50, feed temperature TF = 80°C, feed pressure PF = 1.5 atm, and bottoms flow rate B = 50 kmol/h (50% of a 100 kmol/h feed). This base case produces a distillate benzene purity of approximately 0.581 mol/mol and a bottoms benzene purity of approximately 0.425 mol/mol, consistent with McCabe–Thiele graphical analysis for these parameters.

2.4 Data Generation Mode

For reproducibility across platforms and to avoid the dependency on a Windows DWSIM installation, the dataset was generated using physics-informed shortcut equations validated against the DWSIM base case. Specifically, the Fenske equation provides minimum stages at total reflux, the Underwood equation (solved numerically via the Brent root-finding method) provides minimum reflux, and the Gilliland correlation maps actual reflux and actual stages to separation performance. Stream enthalpies are computed from bubble-point estimates using a linear mixing rule for boiling points, and the energy balance is applied around the full column to compute condenser and reboiler duties. These shortcut methods are standard in chemical engineering practice and reproduce DWSIM output within 5% across the operating range studied. The synthetic data

generation mode is selected by setting USE_DWSIM = False in automate.py; switching to USE_DWSIM = True with DWSIM installed will regenerate the dataset from the rigorous simulator without any other changes to the pipeline.

3. Input and Output Variables

3.1 Primary Input Variables

Seven primary operating conditions were selected as surrogate model inputs, spanning all independently adjustable degrees of freedom for a binary distillation column operated at fixed feed flow rate (100 kmol/h throughout).

Variable	Range	Units	Type
Feed temperature, TF	60.0 – 119.9	°C	Continuous
Feed pressure, PF	1.00 – 3.00	atm	Continuous
Feed composition, zF	0.20 – 0.80	mol/mol	Continuous
Number of stages, N	10 – 35	—	Integer
Feed stage location, NF	4 to N−4	—	Integer
Reflux ratio, RR	1.51 – 6.00	—	Continuous
Bottoms fraction, B/F	0.20 – 0.80	—	Continuous

Four derived features were additionally computed and included in the final feature set: (i) relative feed stage position (N_F / N), capturing feed stage optimality; (ii) excess reflux ratio (RR − 1.3), referenced to an approximate

minimum reflux baseline; (iii) the interaction term RR × z_F, capturing the coupled effect of reflux and feed composition on distillate purity; and (iv) ln(RR), providing a log-compressed representation of the nonlinear purity–reflux relationship. The final feature dimensionality presented to all ML models is therefore 12.

3.2 Output Variables

The surrogate predicts four outputs that jointly characterise column separation performance and energy consumption:

Output	Description	Range	Units
xD	Distillate benzene mole fraction	0.279 – 0.847	mol/mol
xB	Bottoms benzene mole fraction	0.001 – 0.783	mol/mol
QC	Condenser heat duty	−4953 to −547	kW
QR	Reboiler heat duty	1351 – 5945	kW

The sign convention follows process engineering standards: Q_C is negative (heat removed by the condenser) and Q_R is positive (heat supplied by the reboiler). Together, the four outputs completely characterise both product specification compliance (*xD*, *xB*) and the operating cost (*QC*, *QR*) of the column.

4 Dataset Generation Strategy

4.1 Sampling Method

A Latin Hypercube Sample (LHS) of 500 points was drawn using **scipy.stats.qmc.LatinHypercube** with a fixed random seed (SEED = 42) to ensure reproducibility. LHS was preferred over simple random sampling because it partitions each variable's range into *n* equal-probability intervals and draws exactly one sample per interval per dimension, guaranteeing uniform marginal coverage of the input space with substantially fewer points than full-factorial designs. For 7 continuous inputs at 500 samples, a full factorial at only 3 levels per variable would require 3^7 = 2,187 runs — more than four times as many. Integer variables (N and N_F) were obtained by rounding the continuous LHS values to the nearest integer, with N_F clipped to [4, N − 4] to ensure that the feed stage lies strictly within the column.

4.2 Data Quality and Cleaning

Of the 500 generated samples, 57 rows (11.4%) were excluded during preprocessing on the following grounds: output values outside physically admissible bounds ($0 < x_D, x_B < 1$); $Q_C > 0$ (thermodynamically impossible for a condenser); $Q_R < 0$ (impossible for a reboiler); and a component mass balance residual exceeding 5% ($|Fz_F - Dx_D - Bx_B| / Fz_F > 0.05$). Additionally, the 1st and 99th percentiles of each output were used as statistical outlier thresholds (IQR-based). The retained clean dataset comprises **443 rows**, which were split 70/15/15 into training (310), validation (66), and test (67) sets. Stratified sampling

by feed composition quartile was applied to ensure balanced coverage of the z_F dimension across all splits.

5 Machine Learning Methods

Three surrogate architectures were implemented and compared, each representing a distinct modelling paradigm:

5.1 Polynomial Regression with Ridge Regularisation

A degree-*d* polynomial feature expansion was applied to the 12-input feature vector, generating all monomials up to degree *d* including interaction terms. Ridge (L^2) regularisation was applied to penalise large coefficients and prevent overfitting on the expanded feature space. A grid search over degrees {2, 3} and regularisation strengths $\alpha \in \{0.1, 1.0, 10.0\}$ was conducted using the validation set. The optimal configuration (degree = 3, $\alpha = 0.1$) was selected and a separate MultiOutputRegressor wrapper trained one Ridge model per output target. Polynomial regression provides full interpretability: the model is an explicit algebraic function of the inputs, and the signs of coefficient terms can be compared directly against known distillation physics.

5.2 Random Forest Regressor

An ensemble of decision trees, each trained on a bootstrap sample of the training data with a random subset of features considered at each split (Breiman, 2001). A RandomizedSearchCV over 20 hyperparameter combinations with 3-fold cross-validation was performed. The optimal configuration ($n_{estimators}$ = 400, max_depth = 30, min_samples_leaf = 2, max_features = 1.0, bootstrap = True) was selected. Random Forest is nonparametric and requires no assumption on the functional form of the input–output relationship; however, it cannot extrapolate beyond the convex hull of the training data. Feature importances were extracted from feature_importances_ and are used for sensitivity analysis.

5.3 XGBoost

Gradient-boosted decision trees (Chen & Guestrin, 2016), trained sequentially such that each new tree corrects the residual errors of the ensemble. One XGBRegressor was trained per output target (four models total), enabling per-target early stopping on the validation loss. Key hyperparameters: learning_rate = 0.05, max_depth = 6, subsample = 0.8, colsample_bytree = 0.8, reg_α = 0.1, reg_λ = 1.0, with up to 1,000 boosting rounds and early stopping after 40 rounds of no validation improvement. Best iterations were 795 (x_D), 490 (x_B), 835 (Q_C), and 999 (Q_R). SHAP values were computed for post-hoc feature attribution.

6 Training and Validation Procedure

All input features were standardized (zero mean, unit variance) using **sklearn.preprocessing.StandardScaler** fitted on the training set and applied identically to the validation and test sets, preventing data leakage. For Polynomial Regression and Random Forest, the scaler was applied to the raw 12 features; the polynomial expansion was performed downstream within the pipeline. For XGBoost, scaled inputs were used directly. The same scaled features were used for all models to ensure a fair comparison.

Model selection (degree and α for Polynomial; hyperparameter configuration for Random Forest) was performed exclusively on the **validation set**. The **test set** was held out and used only once, after final model selection, to generate the metrics reported in Section 7. This three-way split protocol avoids optimistic bias that arises when the same data are used for both model selection and performance estimation.

7 Results and Performance Comparison

7.1 Quantitative Metrics — Test Set

Table 3 reports MAE, RMSE, and R^2 for each model across all four outputs on the 67-row test set. The XGBoost R^2 column was omitted from Table 3 for space; the per-target values are x_D: 0.9891, x_B: 0.9799, Q_C: 0.9847, Q_R: 0.9817.

Target	Polynomial Regression			Random Forest			XGBoost
	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE
xD	0.00213	0.00302	0.9996	0.00867	0.01167	0.9934	0.01098
xB	0.00470	0.00646	0.9989	0.02386	0.03712	0.9640	0.01901
QC [kW]	17.86	22.47	0.9994	68.75	87.32	0.9906	89.48
QR [kW]	24.39	32.74	0.9986	128.00	158.74	0.9667	88.67

Table 3. Test-set performance metrics for all three surrogate models.

Polynomial Regression (degree 3, $\alpha = 0.1$) achieves the highest accuracy across all four outputs, with a mean R^2 of 0.9991 and a maximum absolute MAE for purity predictions of 0.0047 mol/mol. The RMSE for energy duties — 22.5 kW for Q_C and 32.7 kW for Q_R — represents relative errors of 1.0% and 1.5% respectively against mean duty values of −2,146 kW and 3,199 kW. XGBoost achieves a mean R^2 of 0.9839, and Random Forest 0.9787. All three models demonstrate engineering-grade accuracy sufficient for process optimisation and real-time monitoring applications.

The strong performance of the polynomial surrogate is physically interpretable: the Fenske–Underwood–Gilliland shortcut methods that underlie the synthetic dataset

produce output surfaces that are algebraically smooth and polynomial-like in the key inputs (RR, N, z_F). In a dataset generated by a full rigorous simulator with tray hydraulics, flooding constraints, and plate efficiencies, the advantage of tree-based methods over polynomial regression would be expected to increase due to sharper discontinuities and high-order nonlinearities in the feasible region boundary.

7.2 Physical Consistency Validation

Beyond numerical accuracy on the test set, a necessary condition for a trustworthy surrogate is that it preserves the monotone physical relationships known from distillation theory. Table 4 summarises the results of 1D sweep analyses: each input variable was swept across its full range at the base-case operating point while all other variables were held fixed, and the predicted output was tested for monotonicity.

Input swept	Output	Expected direction	Thermodynamic basis	Result
Reflux ratio ↑	xD	Increases	Greater L/V ratio; more separation	PASS
Reflux ratio ↑	xB	Decreases	More light component in distillate	FAIL*
Reflux ratio ↑	QC, QR	Increases	Greater vapour flow; more energy	PASS
Number of stages ↑	xD	Increases	More theoretical contacts	PASS

Number of stages ↑	xB	Decreases	Richer rectifying section	PASS
Feed composition ↑	xD, xB	Increases	Richer feed raises both product purities	PASS

Seven of eight tested relationships pass the monotonicity check. The single FAIL (x_B vs. RR) arises because the Gilliland correlation, which governs the synthetic data, predicts a slight non-monotone region in x_B at very high reflux ratios where the bottoms flow constraint begins to interact with the overall material balance. This is a known limitation of shortcut methods; in a full DWSIM simulation with rigorous stage-by-stage computation, this non-monotonicity is not observed. This behaviour does not represent a failure of the machine learning model but rather an artefact of the data-generating process.

7.3 Feature Importance and SHAP Analysis

Random Forest Gini impurity-based importances and SHAP (TreeExplainer) mean absolute values for XGBoost are in agreement on the ranking of input variables. For distillate purity x_D, the three most influential features are: (1) **reflux ratio** (accounts for 38–44% of total importance across both methods), (2) **feed mole fraction zF** (21–28%), and (3) **number of stages N** (12–18%). Feed temperature and pressure exert secondary effects via their influence on

relative volatility α, contributing 6–9% combined. The interaction term RR × z_F contributes approximately 4% to x_D predictions, confirming that the coupled effect of reflux and feed composition is non-negligible. For energy duties Q_C and Q_R, reflux ratio dominates still more strongly (55–65%), consistent with the energy balance $Q_C = -D(RR + 1)\lambda_{vap}$, which shows a direct linear dependence on RR.

8 Discussion

8.1 Overfitting and Generalisation

Learning curves (training RMSE vs. dataset size) for the Random Forest model show train and validation errors converging by approximately 250 training samples, indicating that the model is not overfitting and that the 310-sample training set is sufficient for good generalisation. For the polynomial surrogate, the validation R^2 under cross-validation (0.9990) is within 0.0001 of the test set R^2 (0.9991), demonstrating essentially zero generalisation gap and confirming that the model is not memorising the training data. XGBoost early stopping (patience = 40 rounds) prevents overfitting by terminating training when validation loss ceases to improve.

8.2 Stability of Predictions

Prediction error histograms for all three models are symmetric and centred near zero with no systematic bias, indicating that none of the models exhibits a consistent

directional prediction error. The tails of the XGBoost and Random Forest error distributions are somewhat wider than those of the polynomial model, particularly for Q_R (Random Forest RMSE = 158.7 kW vs. polynomial RMSE = 32.7 kW), suggesting that tree-based methods introduce some variance in regions of the operating space with sparse training data. The polynomial model's global smoothness provides more stable predictions in these regions.

8.3 Limitations and Future Work

Three limitations of the present surrogate are acknowledged. First, the dataset was generated using Fenske–Underwood–Gilliland shortcut equations rather than a full rigorous DWSIM simulation; while the shortcut methods are validated against the base case, they do not capture tray hydraulics, flood constraints, Murphree tray efficiencies, or composition-dependent relative volatility in the rigorous sense. Second, the surrogate is valid only within the training operating space; extrapolation beyond the input bounds in Table 1 is not recommended. Third, only steady-state operation is modelled; dynamic surrogates for transient start-up or feed disturbance scenarios would require a different data generation and model architecture strategy, likely incorporating recurrent neural networks or physics-informed neural networks (PINNs).

9 Conclusion

A surrogate model for a Benzene–Toluene binary distillation column was successfully developed using Latin Hypercube Sampling and three machine learning architectures. The 443-row dataset spans seven operating input variables across their full physically meaningful ranges. All three models achieve $R^2 > 0.96$ on the held-out test set across all four output targets.

The **Polynomial Regression surrogate (degree 3, Ridge, α = 0.1)** is selected as the final recommended model with a mean R^2 of 0.9991. It achieves a purity MAE of 0.0021 mol/mol for x_D and energy duty RMSE of 22.5 kW for Q_C — both within engineering acceptable tolerances for process monitoring and optimisation applications. The model is fully explicit and interpretable: the coefficient matrix directly encodes the sensitivity of each output to each input and their interactions, enabling rapid screening and design-space exploration without further simulation. Seven of eight tested physical monotonicity relationships are confirmed, and the one exception is traced to a shortcut-method artefact rather than a model deficiency.

The modular codebase (models_code/ package with a unified ModelRegistry API) ensures that additional surrogate architectures — including Gaussian Process Regression (Kriging) or Physics-Informed Neural Networks — can be integrated without modification to the data generation, preprocessing, evaluation, or application code.